

PILA, Poland

WMO: 122300

Lat: 53.1304N

Lon: 16.7472E

Elev: 73

StdP: 100.45

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.1	-17.5	0.8	-15.2	-14.3	1.1	-11.4	7.5	5.3	6.6	3.2	1.3	60	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.6	19.1	27.7	18.4	26.0	17.7	20.6	26.8	19.7	25.3	18.9	23.7	3.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	22.3	17.8	12.9	21.7	16.9	12.2	21.0	59.6	26.8	56.7	25.3	53.8	23.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.4	5.6	5.1	DB	-19.1	32.6	4.1	1.9	-22.0	33.9	-24.4	35.0	-26.6	36.1	-29.6	37.4	
			WB	-19.0	22.2	4.2	0.9	-22.0	22.8	-24.5	23.4	-26.8	23.9	-29.9	24.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-2.0	-0.2	2.9	8.4	13.0	16.2	18.7	17.9	13.3	8.6	4.0	-0.4	(6)
(7)		DBStd	8.19	5.46	4.83	4.13	4.11	3.93	3.38	3.35	3.08	3.09	3.77	3.43	4.98	(7)
(8)		HDD10.0	1551	373	285	221	77	14	1	0	0	6	72	182	322	(8)
(9)		HDD18.3	3727	631	518	478	298	170	82	38	47	154	302	431	579	(9)
(10)		CDD10.0	971	0	0	1	29	108	186	268	244	104	28	1	0	(10)
(11)		CDD18.3	106	0	0	0	1	6	17	48	33	2	0	0	0	(11)
(12)		CDH23.3	1113	0	0	0	17	83	210	491	278	34	0	0	0	(12)
(13)		CDH26.7	273	0	0	0	2	12	49	152	54	4	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.5	2.6	2.7	2.5	2.4	2.5	2.3	2.1	2.1	2.3	2.4	(14)	
(15)	Precipitation	PrecAvg	603	46	35	43	32	59	63	77	70	50	39	41	48	(15)
(16)		PrecMax	785	99	85	95	86	134	126	154	151	133	112	107	96	(16)
(17)		PrecMin	387	3	7	9	4	22	22	20	11	10	12	2	9	(17)
(18)		PrecStd	102	25	17	22	19	28	27	36	33	24	24	20	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	11.4	16.9	25.6	28.2	30.9	32.5	30.8	27.0	20.4	13.3	11.2	(19)
MCWB			8.3	8.9	10.1	15.0	18.0	21.0	20.4	19.8	17.6	16.0	11.8	10.1	(20)	
2%		DB	8.0	9.1	14.0	21.9	25.8	27.9	30.3	28.1	23.8	18.2	11.5	8.7	(21)	
		MCWB	6.9	7.4	8.5	13.1	16.7	18.0	19.3	18.9	16.5	14.5	10.1	7.6	(22)	
5%		DB	6.4	7.6	11.3	18.7	23.4	25.7	28.2	26.5	21.6	16.1	10.1	7.0	(23)	
		MCWB	5.4	6.1	8.1	11.4	15.7	17.4	18.8	18.3	15.7	13.5	8.8	6.1	(24)	
10%		DB	5.0	6.1	9.2	16.4	21.2	23.5	26.2	24.6	19.5	14.5	8.9	5.7	(25)	
		MCWB	4.1	4.6	6.7	10.4	14.2	16.3	18.1	17.6	14.7	12.4	7.9	4.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	9.4	11.2	15.6	19.3	21.2	21.8	21.2	18.9	16.9	12.2	10.5	(27)
MCDWB			10.0	10.7	14.6	24.0	26.0	27.9	29.0	27.2	24.2	19.8	12.9	11.1	(28)	
2%		WB	7.0	7.8	9.7	13.9	17.7	19.6	20.7	20.3	17.5	15.0	10.3	7.6	(29)	
		MCDWB	7.9	8.9	12.6	20.5	23.7	25.5	27.4	25.6	22.2	17.2	11.3	8.5	(30)	
5%		WB	5.6	6.3	8.1	12.3	16.4	18.2	20.0	19.4	16.4	13.7	9.0	6.2	(31)	
		MCDWB	6.4	7.4	10.7	17.2	22.0	23.9	26.1	24.0	20.1	15.6	9.8	6.9	(32)	
10%		WB	4.1	4.8	6.9	11.0	15.2	17.2	19.0	18.6	15.4	12.6	8.0	5.0	(33)	
		MCDWB	4.9	5.9	9.2	15.3	19.9	21.8	24.1	22.8	18.6	14.3	8.8	5.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	5.8	7.9	10.8	11.0	10.6	10.0	10.4	9.4	8.0	4.9	4.3	(35)
MCDBR			5.0	7.0	12.3	15.1	14.6	14.2	13.8	13.4	13.7	10.1	6.2	5.0	(36)	
MCWBR			4.6	5.5	7.7	7.6	6.9	6.3	4.9	5.7	7.1	6.8	5.1	4.2	(37)	
5% WB		MCDWR	5.0	6.5	10.3	13.0	12.6	12.3	11.5	10.9	11.7	9.2	6.0	4.7	(38)	
		MCWBR	4.7	5.5	7.0	7.0	6.5	6.1	4.9	5.4	6.6	6.5	5.2	4.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.338	0.372	0.412	0.409	0.404	0.430	0.426	0.383	0.364	0.342	0.307	(40)	
(41)		taud	2.295	2.296	2.255	2.203	2.263	2.302	2.246	2.264	2.362	2.391	2.355	2.313	(41)	
(42)		Ebn at Noon	658	756	804	816	840	848	815	795	788	719	615	588	(42)	
(43)		Edh at Noon	62	84	107	127	126	123	128	119	96	76	59	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.62	1.31	2.49	4.14	5.12	5.37	4.99	4.34	3.15	1.70	0.69	0.43	(44)	
(45)		RadStd	0.10	0.20	0.35	0.55	0.57	0.46	0.64	0.42	0.38	0.25	0.10	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-150	+65	N/A

Nomenclature: See separate page